**4.16 use average bond energies to calculate the enthalpy change during a simple chemical reaction.**

## c) Rates of reaction

*Students will be assessed on their ability to:*

4.17 describe experiments to investigate the effects of changes in surface area of a solid, concentration of solutions, temperature and the use of a catalyst on the rate of a reaction

4.18 describe the effects of changes in surface area of a solid, concentration of solutions, pressure of gases, temperature and the use of a catalyst on the rate of a reaction

4.19 understand the term activation energy and represent it on a reaction profile

4.20 explain the effects of changes in surface area of a solid, concentration of solutions, pressure of gases and temperature on the rate of a reaction in terms of particle collision theory

4.21 understand that a catalyst speeds up a reaction by providing an alternative pathway with lower activation energy.

## d) Equilibria

*Students will be assessed on their ability to:*

4.22 recall that some reactions are reversible and are indicated by the symbol $\rightleftharpoons$ in equations

4.23 describe reversible reactions such as the dehydration of hydrated copper(II) sulfate and the effect of heat on ammonium chloride

4.24 understand the concept of dynamic equilibrium

4.25 predict the effects of changing the pressure and temperature on the equilibrium position in reversible reactions.